OussamaLabs

TC Cure Engine

Thermocouple cure analysis for tyres & rubber

PCR/LTR · 22ch · v6.0

demo.operator OPERATOR Log out

Thermocouple Cure Analysis — **Cure Test Report** · SUV 235/60 R18 · cycle trial -45 s

1 TEST IDENTITY

Who / what / when, cure conditions, and the resolved cure-event timing with the path that produced each timestamp.

TYRE SUV 235/60 R18 · cycle trial -45 s	TREAD COMPOUND TR-SUV	SCAN INTERVAL 5s	CHANNELS 22 of 22 usable
CURE START 240s Manual input (240s)	PRESS OPEN 855s Manual input (615s)	DEFLATION 865s Manual input (625s)	TOTAL CURE 630s Manual input (630s)
END OF RECORDING 2395s	TIMING Standard all four entered	ANALYSED BY demo.operator operator	EA SOURCE Compound DNA DNA v1

- **Standard timing.** Cure start, HPS hold, deflation and total cure were all entered; the thermocouple cross-check agrees within 30 s.

TIMING CROSS-CHECK	ENTERED	SEEN IN TC DATA	GAP	STATUS	DETECTOR
Cure start	240s	225s	+15s	OK	Auto: median of 6 sidewall ch
Deflation	865s	855s	-10s	OK	Auto: 3 drops after peak (Ch12 ILM)

Advisory only: entered timing always drives the analysis. A gap beyond 30 s means the stopwatch record and the thermocouples disagree and should be checked. For reference, on the reference validation test the automatic detectors were within -5 s at cure start and +20 s at deflation of the recorded times.

CURE CONDITIONS (documentation)

OPERATOR S. Okafor	PRESS # P-07	MOULD # M-2360-01	BLADDER BL-18-A (95 cycles)
STEAM (BAR) 15.5	PLATEN °C 172	STEAM °C 198	REF Trial CT-18-045

2 PRESS HEALTH — TOP/BOTTOM DELTA @ DEFLATION

Symmetry across the 9 instrumented top/bottom pairs. A **trustworthiness gate**, not a cure verdict — large deltas point to press asymmetry.
Advisory only: no delta changes the cure judgement.

ZONE	PAIR	COMPOUND	TOP°C	BOT°C	ΔT	STATUS	NOTE
Tread Shoulder	TST/TSB	TR-SUV	170.5	169.8	0.7	PASS	within tolerance
Belt Edge	SB2T/SB2B	BC-01	170.6	170.1	0.5	PASS	within tolerance
Sidewall Upper	SWTMT/SWTMB	SW-01	170.8	170.4	0.4	PASS	within tolerance
Sidewall Middle	SWUMT/SWUMB	SW-01	172.4	172.1	0.3	PASS	within tolerance
Mud Stopper	SWULT/SWULB	RC-01	168.2	167.9	0.3	PASS	within tolerance
Apex Inside	AIT/AIB	AP-01	169.3	169.1	0.2	PASS	within tolerance
Apex/Bead	A/BT/A/BB	AP-01	166.7	166.3	0.4	PASS	within tolerance
Inner Liner	ILT/ILB	IL-01	173.2	173	0.3	PASS	within tolerance
Under Belt	UBT/UBB	UB-01	173.1	172.8	0.3	PASS	within tolerance

- **All pairs within tolerance.** Press hardware OK.

3 | CURE VERDICT

OVERALL RESULT

PASS

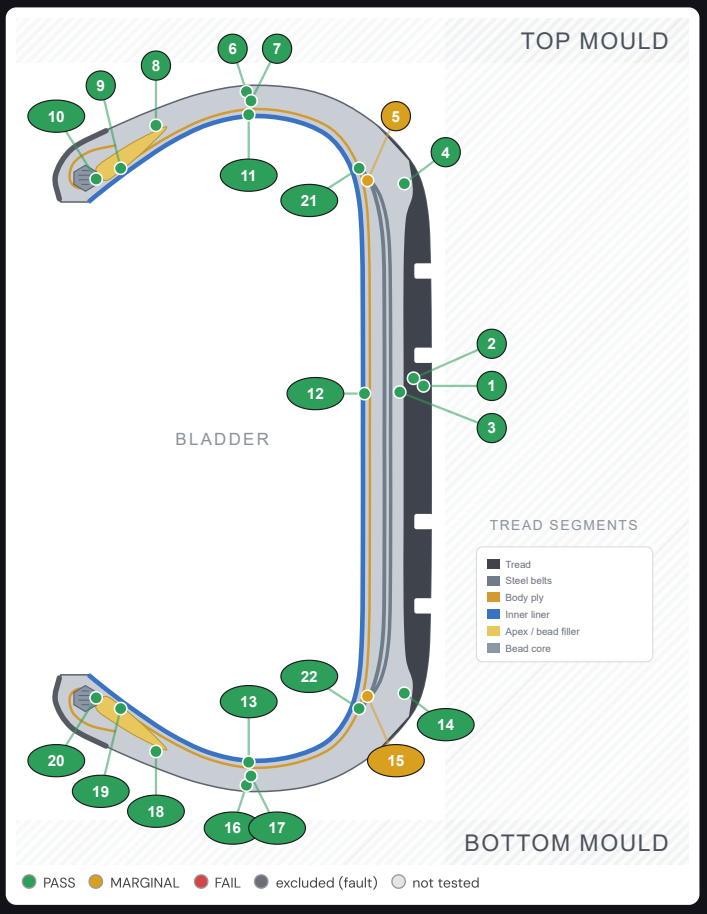
ran hot at some positions, within the compound window — no action required.

20 PASS · 2 MARGINAL · 0 FAIL · none under-cured.

How to read the verdict: PASS means every position is acceptable. MARGINAL means a position ran above its temperature limit but stayed inside the rubber's safe cure window, so no action is needed. FAIL means a position is under-cured or overheated beyond its safe window, and R&D must review the test.

SECTION MAP

Each thermocouple position on the tyre half-section (press orientation, top and bottom mould), coloured by its result. Hover a number for the detail.

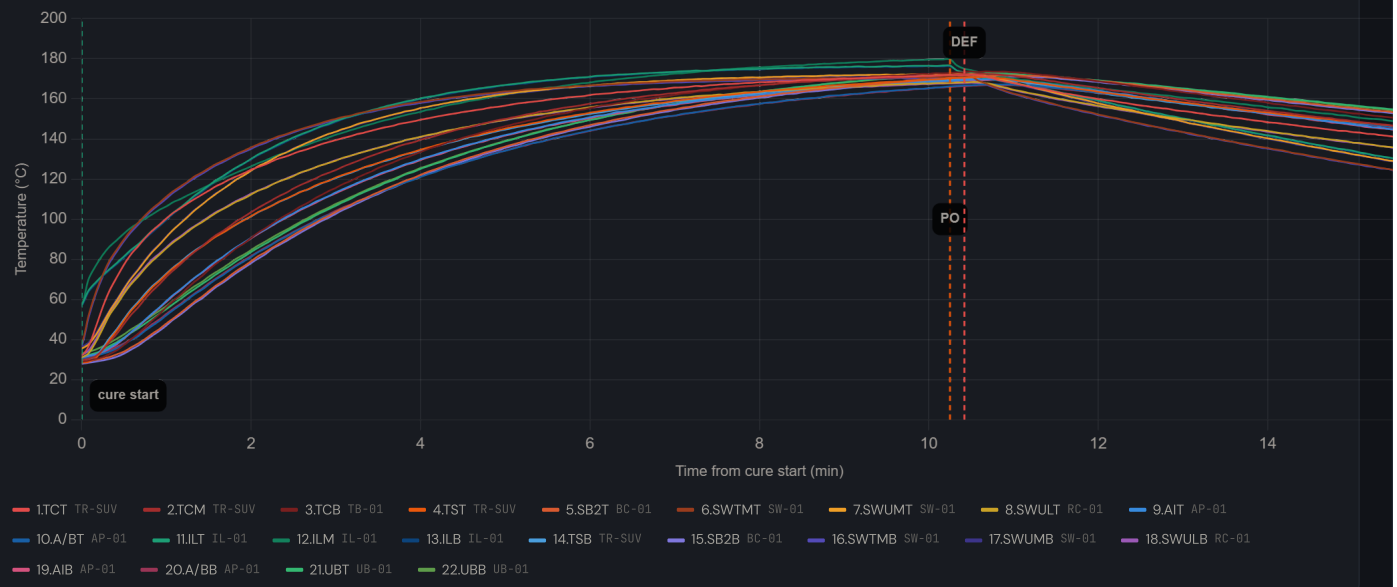


CURE PROFILE CHARTS

All 22 usable positions, shared time axis (minutes from cure start). Same colour per position across charts; click a legend entry to isolate. Dashed markers: cure start, PO, DEF.

1 · Temperature Profile

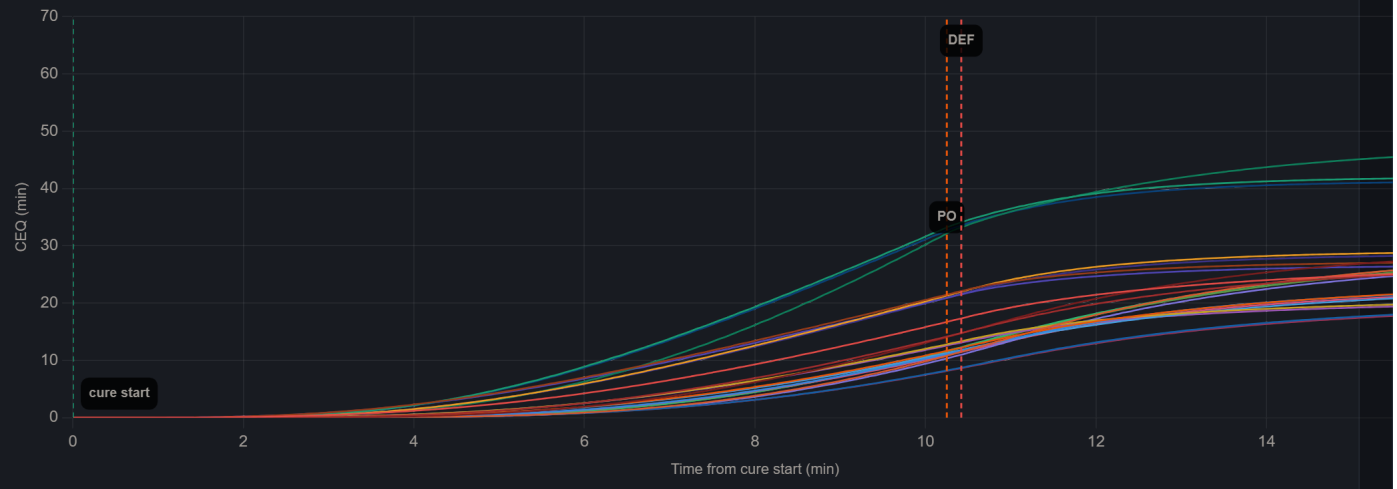
Measured thermocouple temperature per position.



1.TCT TR-SUV · 2.TCM TR-SUV · 3.TCB TB-01 · 4.TST TR-SUV · 5.SB2T BC-01 · 6.SWTMT SW-01 · 7.SWUMT SW-01 · 8.SWULT RC-01 · 9.AIT AP-01
10.A/BT AP-01 · 11.ILT IL-01 · 12.ILM IL-01 · 13.ILB IL-01 · 14.TSB TR-SUV · 15.SB2B BC-01 · 16.SWTMB SW-01 · 17.SWUMB SW-01 · 18.SWULB RC-01
19.AIB AP-01 · 20.A/BB AP-01 · 21.UBT UB-01 · 22.UBB UB-01

2 · CEQ Accumulation

Cumulative cure-equivalent minutes (Arrhenius, 150 °C ref) per position.



1.TCT TR-SUV · 2.TCM TR-SUV · 3.TCB TB-01 · 4.TST TR-SUV · 5.SB2T BC-01 · 6.SWTMT SW-01 · 7.SWUMT SW-01 · 8.SWULT RC-01 · 9.AIT AP-01
10.A/BT AP-01 · 11.ILT IL-01 · 12.ILM IL-01 · 13.ILB IL-01 · 14.TSB TR-SUV · 15.SB2B BC-01 · 16.SWTMB SW-01 · 17.SWUMB SW-01 · 18.SWULB RC-01
19.AIB AP-01 · 20.A/BB AP-01 · 21.UBT UB-01 · 22.UBB UB-01

4 RESULTS PER POSITION — CEQ

CEQ = cure-equivalent minutes at 150 °C accumulated at each position. It is judged at press-open and at deflation; CEQ@End includes the cure gained while cooling.

CH	POSITION	TMAX °C	CEQ@PO	CEQ@DEF	CEQ@END	VERDICT	COMMENT
1	TCT TR-SUV	172	16.72	17.32	27.14	PASS	—
2	TCM TR-SUV	172.4	14.17	14.78	27.78	PASS	—
3	TCB TB-01	173.6	14.05	14.73	30.76	PASS	—
4	TST TR-SUV	170.9	11.68	12.23	24.79	PASS	—
5	SB2T BC-01	171.7	10.81	11.48	30.08	MARGINAL	Over-temperature, within the reversion window
6	SWTMT SW-01	171.1	21.48	22.09	27.41	PASS	—
7	SWUMT SW-01	172.5	21.3	21.97	29.23	PASS	—
8	SWULT RC-01	168.4	12.81	13.34	20.83	PASS	—
9	AIT AP-01	169.8	11.12	11.69	23.57	PASS	—
10	A/BT AP-01	167.3	8.24	8.73	20.48	PASS	—
11	ILT IL-01	176.6	33.22	34.1	42.14	PASS	—
12	ILM IL-01	179.8	32.18	33.2	48.19	PASS	—
13	ILB IL-01	176.3	32.68	33.56	41.44	PASS	—
14	TSB TR-SUV	170.2	11.28	11.82	23.99	PASS	—
15	SB2B BC-01	171.1	10.35	11	28.93	MARGINAL	Over-temperature, within the reversion window
16	SWTMB SW-01	170.7	20.91	21.51	26.73	PASS	—
17	SWUMB SW-01	172.1	20.94	21.6	28.71	PASS	—
18	SWULB RC-01	168	12.59	13.11	20.47	PASS	—
19	AIB AP-01	169.3	10.92	11.48	23.15	PASS	—
20	A/BB AP-01	167	8.17	8.64	20.2	PASS	—
21	UBT UB-01	173.2	11.64	12.31	30.97	PASS	—
22	UBB UB-01	172.9	11.55	12.21	30.48	PASS	—

7 SIGN-OFF

Verification and approval for this thermocouple cure analysis.

Section cut performed: ..... All TC locations verified: .....

Tester: ..... Approved by: .....

Comments: .....